

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Generator Step-up Transformers - Datasheet

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Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
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	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
Signature / date	signed 2026-09-27	signed 2026-09-27	signed 2026-09-27

Rev	Date	Description
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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Generator step-up transformers

Item	GT GSU	ST GSU
Rating	560 MVA ONAF (336 MVA ONAN)	270 MVA ONAF (162 MVA ONAN)
Voltage ratio / vector group	400 +/- 8 x 1.25 % / 21 kV, YNd11, OLTC	400 +/- 8 x 1.25 % / 15.75 kV, YNd11, OLTC
Impedance	14.5 %	13.5 %
No-load losses / load losses at rating	190 kW / 1,150 kW	110 kW / 620 kW
Losses at SRC-NG-100 load (P, pf 0.85)	1,110 kW	580 kW
Insulation level HV / neutral	LI 1425 kV / SI 1050 kV; neutral solidly earthed	same
Oil	natural ester (K-class) - fire risk reduction; oil volume 98 m3	natural ester; 62 m3
Transport weight (without oil, nitrogen-filled) / dimensions	268 t / 11.2 x 4.1 x 4.6 m	168 t / 9.4 x 3.8 x 4.3 m
Total weight	395 t	245 t
HV connection	HV bushings with air terminal; 380 kV connection to the GIS by EPC (IF-34/IF-35)	same
Monitoring	on-line DGA, bushing monitoring, fibre-optic hot-spot; IEC 61850 to the EPC substation system	same

2. Interfaces

- Transformer foundations, oil pits (110 % oil volume), fire walls and deluge system: by EPC (NFPA 850, IEC 61936-1).
- 380 kV HV connection from the bushings to the GIS, surge arresters at the HV bushings: by EPC.
- IPB connection flanges at the LV bushings: by IEC.